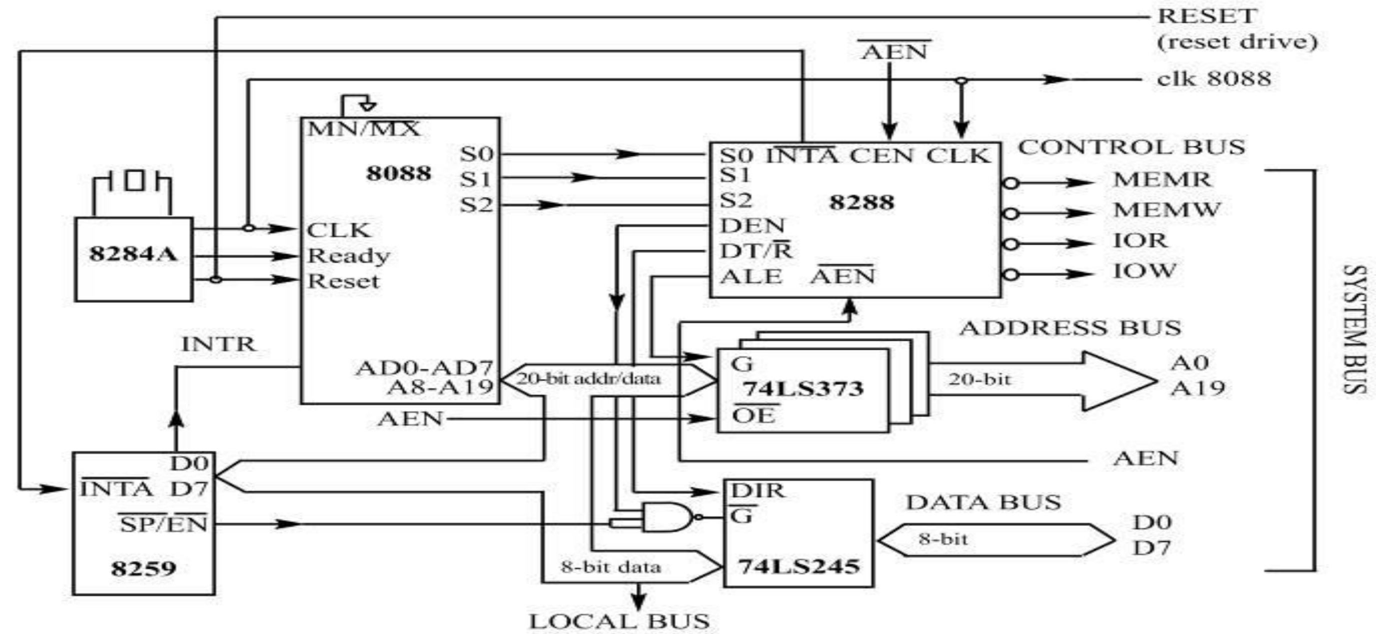
Arab Academy for Science, Technology & Maritime Transport
College of Engineering and Technology
Final Examination Paper

Department: Computer Engineering **Date**: 7 June 2020
Course Title: Introduction to Microprocessors. **Time** : **2** Hours
Course Code: CC411 **Marks**: 40
Lecturer: Dr. Cherine Fathy

FINAL EXAM

Question #1 (10 Marks) (ILO A1, B1, Outcome 1)



A. (5 Points) Referring to the above figure of the 8088 CPU and its supporting Chips, answer the following questions:

a. The 8088 CPU is in which mode according to the above figure?

b. When DT/$\overline{R}$ is low and DEN is low, the data buses are used by local bus or System Bus?

c. Which chips are used to isolate the local bus and system bus?

d. At which T state ALE is Disabled?

B. (5 Points) Indicate whether the statement is true or false and <u>correct the false ones</u>

a. Mask ROM is a user-programmable ROM.

b. Read cycle time is measured as time elapsed from the moment an address is provided to the address pins to the moment data is available at the pins.

c. Organizations for DRAMs are always x 8.

d. In minimum mode, 8088 requires the use of the 8288 to generate some of the control signals.

e. The reset signal is generated by the 8288 chip.

Question #2 (10 Marks) (ILO A1, B1, Outcome 1)

A. (6 Points) Using the below memory locations, answer questions 1 to 6

Physical address	Contents
12340 H	12H
12341H	34H
12342H	FAH
12343H	37H
12344H	38H
12345H	39H
12346H	40H
12347H	41H
12348H	00H
12349H	D0H
1234AH	E8H
1234BH	C3H
1234CH	8EH
1234DH	C3H
1234EH	01H
1234FH	20H
12350H	11H
12351H	FFH
12352H	94H

1. If SP = 2344H is used to point to the memory location whose physical address is 12344H, the SS register has……
   a. 1000H
   b. 1010H
   c. 2000H
   d. 2010H

2. A character is stored in its ASCII format starting from the physical address 12344H, its ASCII code is…….
   a. 3837H
   b. 3938H
   c. 38H
   d. 41403938H

3. If SP is initialized at 0000H and SP = 0350H points to the memory location whose address is 12350H, the stack is storing …..words
   a. 424
   b. 848
   c. 32344
   d. 64688
4. According to question 3, the range of the stack segment is…
   a. 03500H-134FFH
   b. 10000H-1FFFFH
   c. 12000H- 21FFFH
   d. 12350H-2234FH
5. According to question 3., after executing POP AX, AX equals… ……….
   a. 11FFH
   b. FF11H
   c. 2011H
   d. 1120H

6. After executing question 5., the SP equals……
   a. 0348H
   b. 0349H
   c. 0351H
   d. 0352H

B. (2 Points) Write an efficient and functionally equivalent instruction for the following instructions:

DEC CX
CMP CX, 0
JNZ label1

C. (2 Points) Indicate the status of ZF and CF after CMP is executed in the following case. Show your work.

SUB CX, CX
DEC CX
CMP CX, 0FFFFH

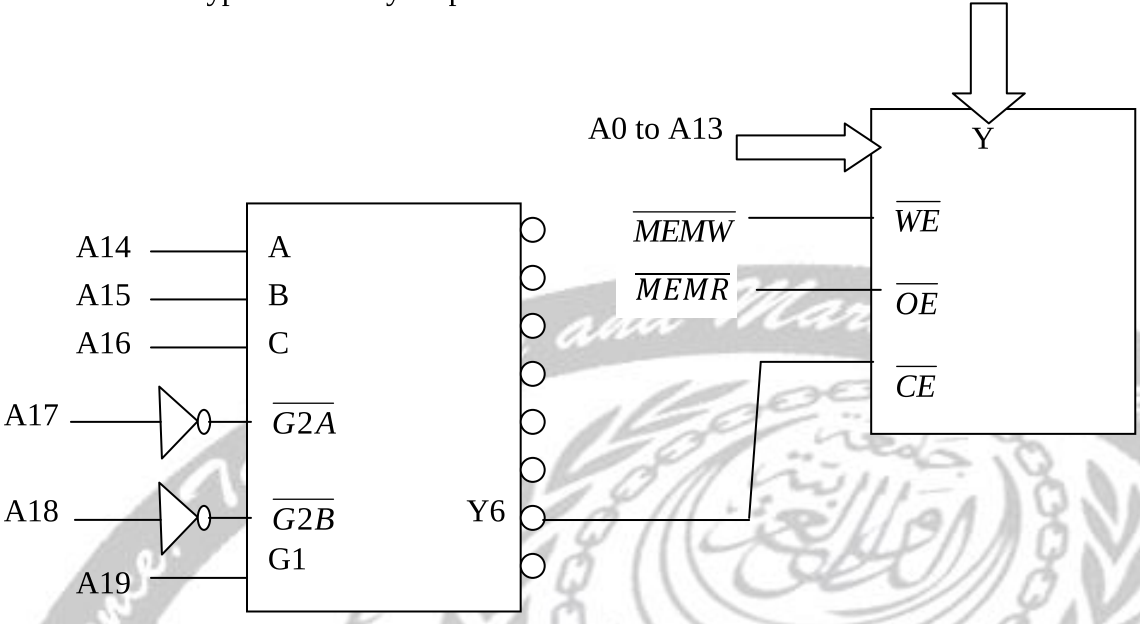
Question #3 (10 Marks) (ILO A1, B1, Outcome 1)

A. (4 Points) For the design shown below, specify the following:

a. The address range for Y6.

b. The memory capacity and organization of the chip selected by Y6.

c. The Type of memory chip: RAM or ROM.

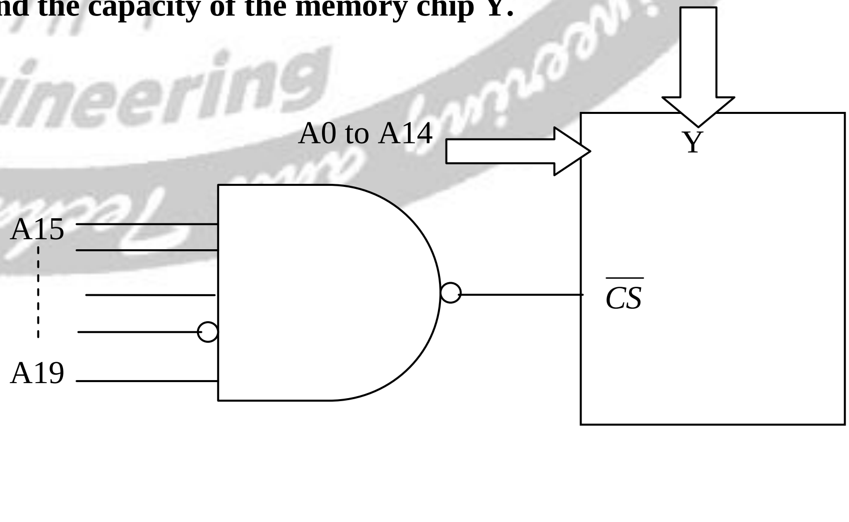


B. (2 Points) What is the main advantage and disadvantage of SRAM memory chips?

C. (4 Points) For the following memory design, Find

a. The address range.

b. The organization and the capacity of the memory chip Y.



Question #4 (10 Marks) (ILO A1, B1, Outcome 1)

A. (5 Points)

```
1.  Program 1
2.  . DATA
3.  NUM      DB    F3H
4.  COUNT  DB    ?
5.  . Code
6.  MAIN PROC FAR
7.  MOV AX, @DATA
8.  MOV DS,AX
9.  MOV CX,8
10. CLC
11. SUB BX,BX
12. MOV AX,NUM
13. BACK: SHR AX,1
14.           JNC NEXT
15.           INC BX ;
16. NEXT: LOOP BACK
17. MOV COUNT,BX
18. MOV AH,4CH
19. INT 21H
20. MAIN ENDP
21. END MAIN
```

i. (2 Points) Explain the function of program 1 and determine the value of count after executing the program.

ii. (1 Point) How many times the loop is executed?

iii. (1 Point) Determine the addressing mode of line 9?

iv. (1 Point) Replace the instruction of line 11 by another instruction that do the same function?

B. (5 Points) X and Y are arrays of 100 Bytes. Write a complete assembly program to perform an exchange of elements of two arrays (swap the values of X and Y).